Thermal effects and energy budgets

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a temperature change only after stating how much energy enters the material.

SUCCESS CRITERIA - I can...

1. Apply an explicit absorbed-energy fraction.
2. Use mass and specific heat in a temperature estimate.
3. State the no-phase-change and no-loss assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “absorbed” and “specific heat” to explain your model or result. Sentence starters are optional.

absorbed: taken into the specified object or system

specific heat: energy per mass needed for a one-degree temperature change

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

Bodies initially at 330 K and 350 K exchange heat in an isolated system.
Explain what happens to total energy and to the temperature-difference opportunity for work at equilibrium.

Word help: heat = energy transferred because of a temperature difference

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q_{absorbed} = mc\,\Delta T$; $\Delta T = Q_{absorbed}/(mc)$.
First identify the absorbed energy. Use only when there is no phase change and c is approximately constant.

Worked reasoning: A surface receives 500 J. A model accounts for 200 J reflected, 250 J absorbed as heating and 50 J transmitted. The full account sums to 500 J.

Your turn: Draw proportional energy arrows and identify which part can directly raise this surface's temperature.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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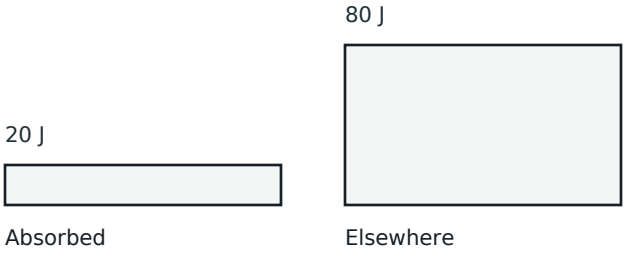
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A model releases 100 J. Only 20% enters a sample.
Draw the energy partition and calculate energy absorbed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: 1 2 3

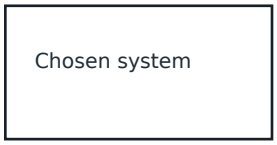
04 Apply the physics

A model energy release totals 2×10^6 J. A 0.50 kg sample absorbs 1%; its specific heat is 800 J/(kg·°C).
Label the energy partition, calculate absorbed energy, then estimate ΔT assuming no phase change or loss.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A temperature prediction exceeds a material's melting point, but the calculation used only mc ΔT. What must be added or reconsidered?

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg cart descends 6 m from rest.
Use g=10; losses are 8 J.
Find bottom speed.
Show the energy changes and transfers; label units.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 2 kg sample with c=500 warms 8°C.
Find absorbed energy. If this is 4% of source energy, find source energy.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.4 kg panel absorbs 1200 J, emits 300 J and does no work. With c=450 J/(kg·K), find internal-energy and temperature changes.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3